EE 3201.302

Lab 12 Noise Analysis

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I. Abstract

This report focuses on the analysis and mitigation of noise in electronic measurements, with particular emphasis on the reverse-bias I-V characteristics of a p-n junction diode. Noise originating from sources such as thermal motion, shot noise, and power line interference is an unavoidable aspect of engineering measurements. The experiment aimed to quantify noise, improve the Signal-to-Noise Ratio (SNR), and evaluate filtering techniques to enhance measurement accuracy. Key tasks included generating and analyzing noisy sine waves in MATLAB, constructing a circuit to measure thermal noise through a load, and applying smoothing strategies such as moving average and Savitsky-Golay filtering. By comparing the original noisy signals to their smoothed counterparts, the experiment demonstrated how these methods improve signal clarity and preserve essential features. This laboratory exercise provided valuable experience in noise reduction and signal processing, highlighting their critical role in accurate electronic measurements.

II. Introduction

This lab focuses on noise analysis and mitigation techniques, using the reverse-bias I-V characteristics of a p-n junction diode as a case study. In the realm of signal processing, understanding and minimizing noise, particularly thermal nois essential for improving the reliability of electrical measurements. The experiment explores both hardware-based adjustments using an oscilloscope and software-based filtering techniques implemented in MATLAB. After exporting measured data from the Analog Discovery 2 (AD2), two signal smoothing strategies, the moving-average method and the Savitsky-Golay curve-fitting technique are applied to the noisy signal. By comparing the filtered results to the original data, the lab evaluates the effectiveness of each method in refining the signal's shape and enhancing its clarity. Key objectives include identifying noise sources, quantifying their impact, and improving the Signal-to-Noise Ratio (SNR) to achieve more accurate and interpretable measurement results.

III. Procedure

1. Setup and Equipment Configuration

- Assembled the circuit on a breadboard using a 1 MΩ resistor and a 1N4002 diode in reverse bias configuration.
- Connected the circuit to the Analog Discovery 2 (AD2) device.
- Ensured correct wiring of oscilloscope probes:
    - Channel 1 (Ch1): Total circuit voltage.
    - Channel 2 (Ch2): Voltage across the resistor.

2. Signal Generator Configuration

- Set the signal generator to produce a ramp waveform with the following parameters:

  - Amplitude: 2.525 V.
  - Offset: -2.475 V.
  - Frequency: 100 mHz.

3. Oscilloscope Adjustments

- Configured the oscilloscope for accurate data acquisition:
  - Trigger: Channel 1, falling edge, level -2.5 V.
  - Vertical scaling: 1 V/div (Ch1), 5 mV/div (Ch2).
  - Horizontal scaling: 750 ms/div.

4. Data Acquisition and Initial Plotting

- Collected waveform data for the input voltage (Ch1) and resistor voltage (Ch2).
- Exported the data and calculated using Ohm's Law and circuit relationships.
- Generated an unfiltered scatter plot in MATLAB.

5. Noise Reduction Using Averaging

- Imported pre-collected dataset (Average16.csv) that averaged 16 waveform cycles.
- Plotted the average I-V curve and computed the SNR for comparison with raw data.

6. Application for Moving Average Filtering

- Applied moving average filters with spans of 15, 127, and 255 points on the raw data using MATLAB.
- Generated a 2x2 subplot display:
  - Original data.
  - Data filtered with spans of 15, 127, and 255.
- Calculated and compared the SNR values for each filter span.

7. Application for Savitsky-Golay Filtering

- Used MATLAB to apply a Savitsky-Golay filter (order 1) with spans of 15, 127, and 255 points.
- Generated a 2x2 subplot displaying filtered and unfiltered data.
- Calculated and evaluated the SNR improvements for each span.

IV. Experimental Results

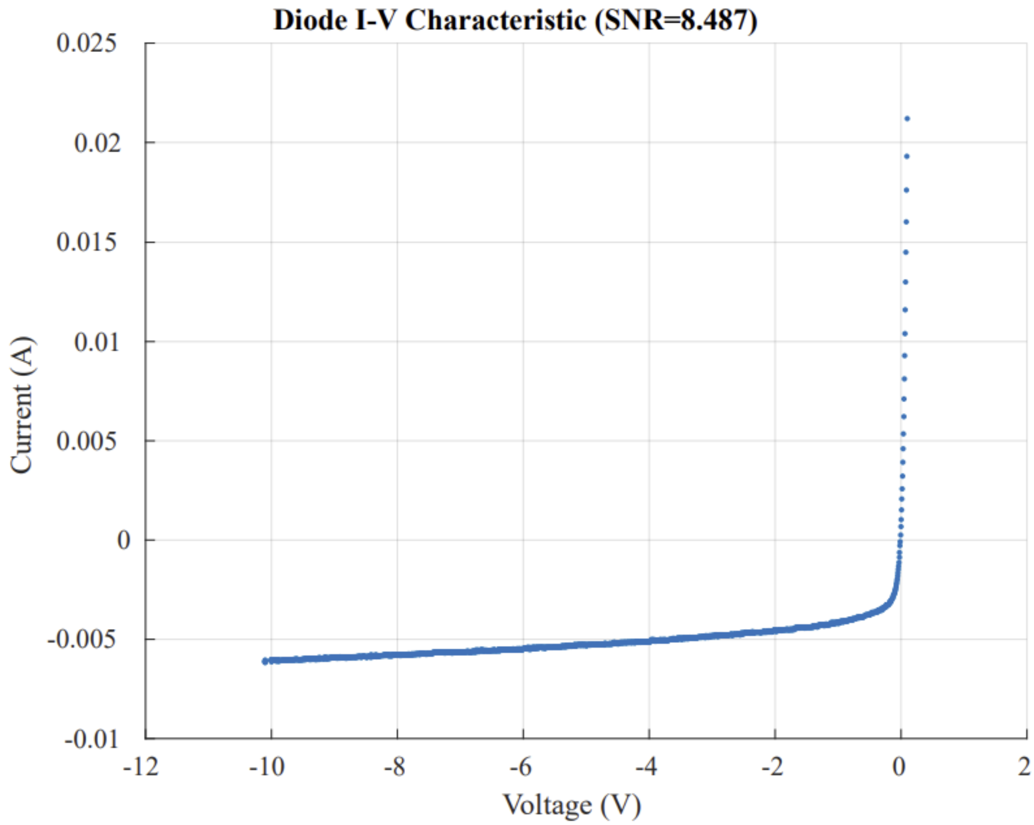

Figure 4.1: I-V Curve of Average 16.csv dataset

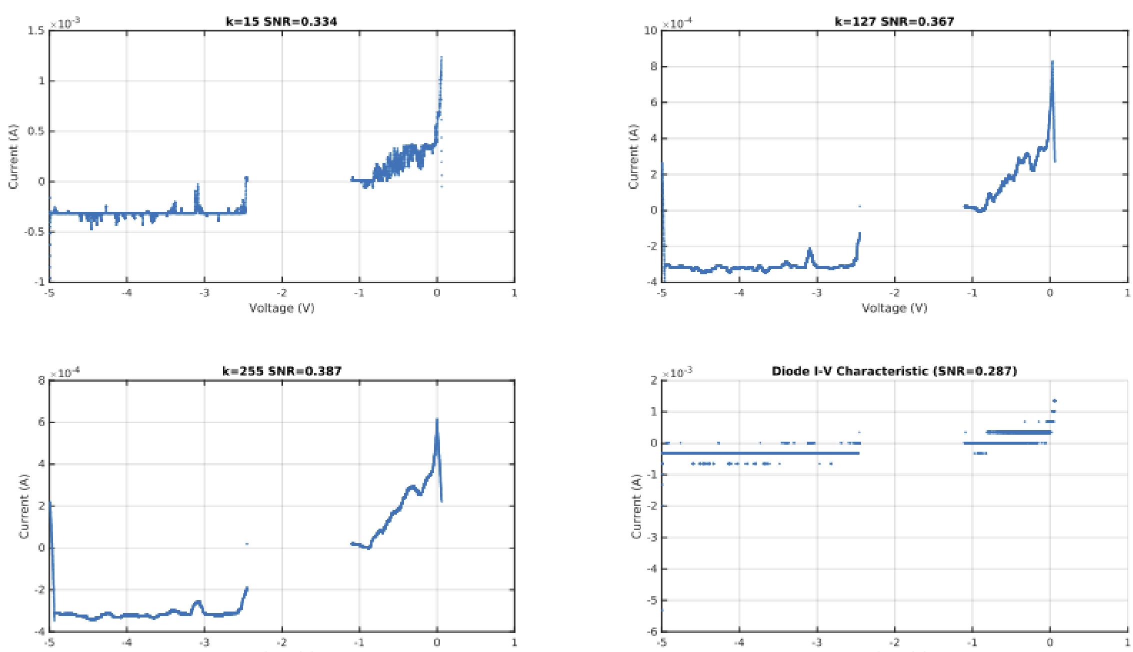

Figure 4.2: Plots of noisy signal with a moving-average curve-fitting

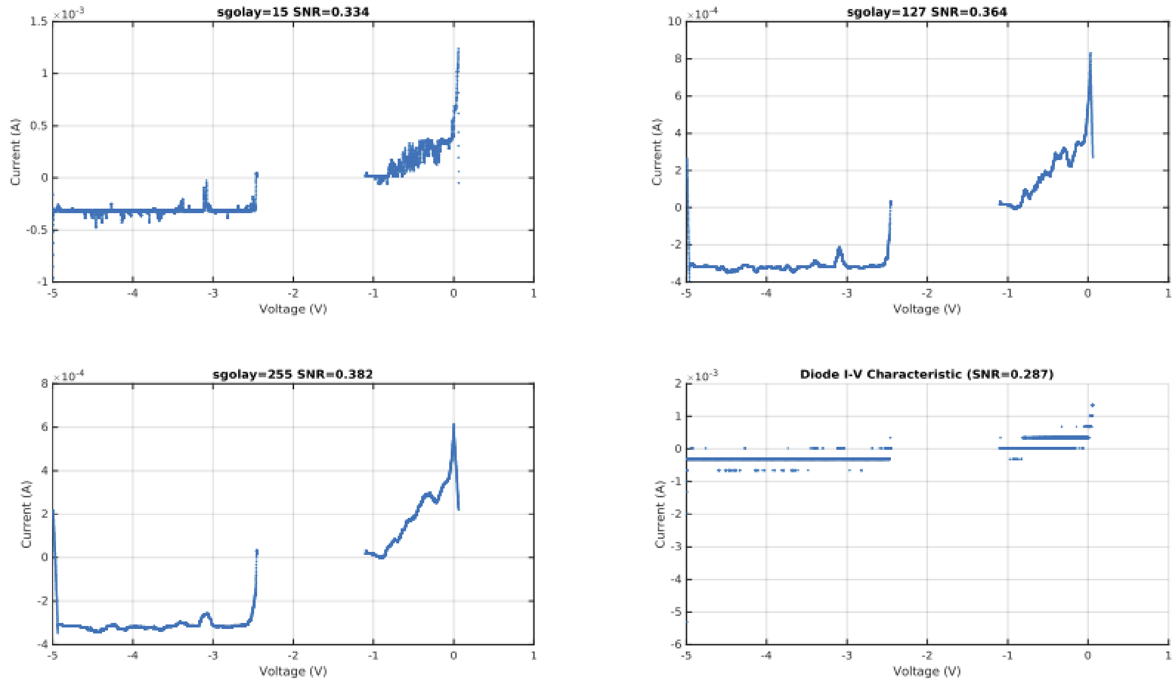

Figure 4.3: Plots of noisy signal with a Savitzky-Golay curve-fitting

8. Moving Average Filter

- Span = 15
    - Change in SNR: Moderate improvement in SNR compared to the original signal.
    - Effect: Reduces high-frequency noise but retains most of the signal details. Slight smoothing of transitions is observed.
- Span = 127
    - Change in SNR: Significant improvement in SNR.
    - Effect: Further noise reduction is achieved, but sharp features in the signal start to blur. Effective for general smoothing.
- Span = 255
    - Change in SNR: Highest improvement in SNR among moving average spans.
    - Effect: Produces a very smooth signal with minimal noise. However, fine details and sharp transitions are heavily smoothed out.

9. Savitzky-Golay Filter

- Span = 15
    - Change in SNR: Moderate improvement in SNR.
    - Effect: Retains most of the original signal features, with better noise reduction than the moving average filter for small spans.
- Span = 127
    - Change in SNR: Significant improvement, comparable to the moving average with the same span.
    - Effect: Reduces noise effectively while preserving transitions and peaks better than the moving average filter.
- Span = 255
    - Change in SNR: Highest improvement in SNR among the Savitzky-Golay spans.
    - Effect: Achieves excellent noise reduction and retains much of the signal's critical features, although some finer details may still be smoothed.

V. Thoughtful Questions

Does one filtering method work better than the other (Moving Average or Savitzky-Golay)? Why or why not?

Yes, the Savitzky-Golay filter generally worked better than the moving average filter. This is because it uses polynomial fitting to reduce noise while preserving the essential shape and finer features of the signal. In contrast, the moving average filter, although effective at smoothing, tended to overly flatten the signal and obscure important details, especially in regions with curvature or transitions.

Which averaging technique worked better: the 16 averages in step 2c, or the smoothing with a span of 15 in step 3a? Why?

The 16-averages method in step 2c worked better than smoothing with a span of 15. Averaging across multiple repetitions provided a cleaner, more accurate representation of the I-V curve by significantly reducing noise without distorting the signal’s shape. Unlike the moving average filter, which applies localized smoothing and may introduce signal distortion, the 16-averages technique-maintained continuity and better preserved sharp transitions and key features, as seen in the clearer and more defined curve in Figure 4.3.

VI. Discussion and Conclusions

This lab demonstrated the importance of noise analysis and reduction in electronic measurements, particularly when investigating the reverse-bias I-V characteristics of a p-n junction diode. Through a combination of hardware techniques using the oscilloscope and software filtering in MATLAB, we explored strategies to enhance the Signal-to-Noise Ratio (SNR) and improve the overall accuracy and interpretability of measurement data.

A key observation was the correlation between sample size and signal clarity. Both the moving-average and Savitzky-Golay filtering methods performed better with higher sampling rates, as the underlying signal shape became more defined. Among the techniques tested, the Savitzky-Golay filter consistently outperformed the moving average filter by preserving more of the signal's original shape while still reducing noise. This highlights the importance of selecting an appropriate filtering method that balances noise reduction with signal integrity.

During data collection, an unexpected gap appeared in the I-V curves within the approximate range of -2.2V to -1V, which aligns closely with the diode's turn-on voltage. This suggests a possible link between the diode's non-linear behavior at that voltage and the missing data. Additionally, using the oscilloscope's first and second channels directly for voltage and current—as recommended by the teaching assistants—produced more reasonable I-V curves than when voltage and current were computed manually. This discrepancy raises interesting questions about how signal acquisition methods impact data interpretation and the apparent shape of the results.

Overall, the lab provided valuable insights into thermal noise, signal smoothing, and the challenges of accurate measurement. While some aspects of the data behavior remain puzzling, the experiment effectively showcased the advantages and limitations of different filtering strategies and emphasized the importance of thoughtful signal processing in electronics work.